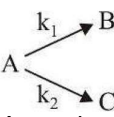


**SCQ (Single Correct Type) :**

- Choose the correct indicator for the following mixture/titration :  
 50ml, 0.1M  $\text{HCO}_3^-$  + 50ml, 0.8 M  $\text{CO}_3^{2-}$  . . .  
 (A) Phenol Red (6.8 – 8.4) (B) Propyl Red (4.6 – 6.4)  
 (C) Phenolphthalein (8.3 – 10.1) (D) Malachite green (11.4 – 13)  
 $\text{H}_2\text{CO}_3$  :  $K_{a1} = 4 \times 10^{-7}$        $K_{a2} = 2 \times 10^{-11}$  . . .  $\log 2 = 0.30$
- 
 $k_1 = x \text{ hr}^{-1}$  ;  $k_1 : k_2 = 1 : 10$ . Calculate  $\frac{[C]}{[A]}$  after one hour from the start of the reaction.  
 Assuming only A was present in the beginning.  
 (A)  $\frac{1}{10}(e^{-11x} - 1)$  (B)  $\frac{10}{11}(e^{-11x} - 1)$   
 (C)  $\frac{10}{11}(e^{11x} - 1)$  (D) None
- The reaction  $2\text{NO} + \text{Br}_2 \longrightarrow 2\text{NOBr}$ , is supposed to follow the following mechanism.  
 (i)  $\text{NO} + \text{Br}_2 \xrightleftharpoons{\text{fast}} \text{NOBr}_2$  (ii)  $\text{NOBr}_2 + \text{NO} \xrightleftharpoons{\text{slow}} 2\text{NOBr}$   
 Suggest the rate of law expression :  
 (A)  $r = K [\text{NO}]^2 [\text{Br}_2]$  (B)  $r = K [\text{NO}][\text{NOBr}]$   
 (C)  $r = K [\text{Br}_2][\text{NOBr}]$  (D)  $r = K [\text{NO}][\text{Br}_2]$
- Titration of  $\text{Na}_3\text{PO}_4$  with HCl to phenolphthalein end point produces  $\text{Na}_2\text{HPO}_4$  ; subsequent titration to methyl orange end point produces  $\text{NaH}_2\text{PO}_4$  . Find the mg of  $\text{Na}_2\text{HPO}_4$  present in a sample which required 25 ml, 0.12 N HCl solution to reach the phenolphthalein end point and an additional 35 ml of the same acid to reach the methyl orange end point.  
 (A) 492 mg (B) 170.4 mg  
 (C) 662 mg (D) Insufficient data is provided

5. A 1 Ltr flask containing  $\text{NH}_3(\text{g})$  at 2.0 atm and 300 K is connected to another 800 ml flask containing  $\text{HCl}(\text{g})$  at 8.0 atm and 300 K by means of a narrow tube of negligible volume and gases were allowed to react quantitatively as  $\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \longrightarrow \text{NH}_4\text{Cl}(\text{s})$ ;  $\Delta H = -43\text{kJ/mol}$ . If heat capacity of  $\text{HCl}(\text{g})$   $C_v$  is  $20\text{J K}^{-1}\text{mol}^{-1}$ , determine final pressure inside the flask assuming negligible heat capacity of flask and  $\text{NH}_4\text{Cl}(\text{s})$ , also negligible volume of solid  $\text{NH}_4\text{Cl}$ .
- (A) 10.4 atm                      (B) 15.6 atm                      (C) 5.4 atm                      (D) 20.3 atm
6. Which of the following statements is incorrect ?
- (A) Caesium is the most electropositive element while F is the most electronegative element  
 (B) Cl has the highest -ve electron gain enthalpy out of all the elements  
 (C) Electrons gain enthalpy of N as well as that of noble gases is positive  
 (D) Electron gain enthalpy of 'F' atom is least negative in it's group
7. Consider the following reactions :
- (I)  $\text{Q}_{(\text{g})} + \text{e}^- \longrightarrow \text{O}_{(\text{g})}^-, \Delta H_1$                       (II)  $\text{F}_{(\text{g})} + \text{e}^- \longrightarrow \text{F}_{(\text{g})}^-, \Delta H_2$   
 (III)  $\text{Cl}_{(\text{g})} + \text{e}^- \longrightarrow \text{Cl}_{(\text{g})}^-, \Delta H_3$                       (IV)  $\text{O}_{(\text{g})}^- + \text{e}^- \longrightarrow \text{O}_{(\text{g})}^{2-}, \Delta H_4 \cdot \cdot \cdot \cdot \cdot \cdot \cdot \cdot$
- Then according to given information the incorrect statement is/are :
- (A)  $\Delta H_3$  is more negative than  $\Delta H_1$  and  $\Delta H_2$   
 (B)  $\Delta H_1$  is less negative than  $\Delta H_2$   
 (C)  $\Delta H_1, \Delta H_2$  and  $\Delta H_3$  are negative whereas  $\Delta H_4$  is positive  
 (D)  $\Delta H_1$  and  $\Delta H_3$  are negative whereas  $\Delta H_2$  and  $\Delta H_4$  are positive

### Comprehension-5

An ore of a common metal contain a green colored compound 'M' of metal, which dissolves readily in hydrochloric acid. However the metal does not dissolve in dilute hydrochloric acid. The compound 'M' gives effervescence of gas 'X' and produces a blue colored solution of 'P' on treatment with HCl. The gas 'X' is non-combustible colorless and odorless. The gas 'X' reacts with aqueous suspension obtained by adding 'Y' in water to give white precipitate of 'Z'. Z on heating gives back X and Y. When gas Q is passed through 'P' a black precipitate of 'r' is obtained. When a gas 'S' is passed through P the blue solution turns deep blue.

8. The compound "M" is :
- (A)  $\text{MgCO}_3, \text{Mg}(\text{OH})_2$                       (B)  $\text{CuFeS}_2$   
 (C)  $\text{CuCO}_3, \text{Cu}(\text{OH})_2$                       (D)  $\text{FeCO}_3$
9. Gases 'Q' and 'S' are allowed to enter a long tube of length 100 cm from opposite ends under identical conditions. What is the distance travelled by 'Q' before it meets 'S'?
- (A) 50 cm                      (B) 41.4 cm                      (C) 46.5 cm                      (D) 59 cm
10. Which of the following is used as raw material in solvay's process to produce  $\text{CO}_2$  ?
- (A) M                      (B) Y                      (C) Z                      (D) R

11. **Statement-1** : MgO bricks are used for inner lining of the zone of combustion of blast furnace in extraction of iron.  
**Statement-2** : MgO fails to react with  $\text{SiO}_2$  under the furnace condition where it is used as inner lining.
12. **Statement-1** : Adding KCN to  $\text{CuSO}_4$  produces a white ppt which dissolves in excess KCN.  
**Statement-2** : With excess KCN,  $\text{CuSO}_4$  forms  $\text{K}_2 [\text{Cu}(\text{CN})_4]$  complex.

**MCQ (Multiple Correct Type) :**

13. A gas of identical H-like atom has some atoms in the lowest (ground) energy level 'A' and some atoms in a particular upper (excited) energy level 'B' and there are no atoms in any other energy level. The excited atoms of the gas make transition to a higher energy level by absorbing monochromatic light of photon energy 2.7 eV. Subsequently, the atoms emit radiation of only six different photons energies. Some of the emitted photons have energy 2.7 eV. Some have more and some have less than 2.7 eV, which of the following statement(s) is/are correct  
**(A)** The principal quantum number of initial excited level 'B' is 2  
**(B)** The ionization energy for gas atoms is 14.4 eV  
**(C)** The minimum energy of emitted photon is 1.4 eV  
**(D)** The minimum energy of emitted photon is 0.7 eV
14. Select the incorrect statement :  
**(A)**  $\text{HSO}_5^-$  ion has one S – O – H linkage  
**(B)** Number of B – O – B linkages in Borax is equal to number of P – O – P linkages in  $\text{P}_4\text{O}_{10}$   
**(C)** Hybridization of both sulphur in  $\text{H}_2\text{S}_2\text{O}_5$  (pyrosulphurous acid) is same but oxidation state of both sulphur are different  
**(D)** Tetra-polyphosphoric acid has four P – O – P and no P – P linkage
15. The correct statement(s) is/are :  
**(A)** Boiling point of m-hydroxybenzaldehyde is greater than o-hydroxybenzaldehyde  
**(B)** Boiling point of  $\text{CHCl}_3$  is higher than  $\text{CCl}_4$   
**(C)** Ionisation energy of  $\text{N}_2 < \text{O}_2$ , but reverse order of ionization energy is true for their corresponding atoms  
**(D)** Boiling point of HF is greater than  $\text{CH}_3\text{F}$
16. Which of the following is/are true?  
**(A)** Due to Schottky defect, density of crystal remains unaffected  
**(B)** In zinc blende,  $\text{S}^{2-}$  forms FCC while alternate tetrahedral voids are occupied by  $\text{Zn}^{2+}$   
**(C)** Number of next nearest neighbours of  $\text{Cs}^+$  ions in CsCl structure is 6  $\text{Cs}^+$  ions  
**(D)** A crystal has A ions at the cube corners and B ions at the body centres. The coordination number of A and B are respectively 8 and 8

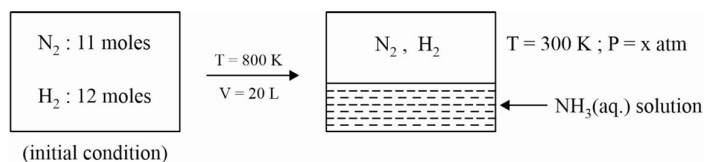
17. 1 mole of a non-volatile non-dissociative solid solute is dissolved in 200 moles of water. The solution is taken to a temperature  $T_K$  (lower than freezing point of solution) to cause ice formation. After removal of ice the remaining solution is taken to 373 K where vapour pressure is observed to be 740 mm of Hg. Identify the correct options: [Given data :  $K_f(\text{H}_2\text{O}) = 2\text{Kkgmol}^{-1}$  and normal boiling point of  $\text{H}_2\text{O} = 373\text{K}$ ]
- (A) 163 moles of ice will be formed
- (B) Temperature to which original solution was cooled should be  $2000 \frac{2000}{37 \times 18} \text{K}$
- (C) Freezing point of original solution should be  $-\frac{10}{18}^\circ\text{C}$
- (D) Relative lowering of vapour pressure of final solution will be  $\frac{1}{201}$
18. For a certain reaction  $A \longrightarrow P$ , the half life for different initial concentration of A is mentioned below.
- |               |     |       |
|---------------|-----|-------|
| [A]           | 0.1 | 0.025 |
| $t_{1/2}$ (s) | 100 | 50    |
- Which of the following options(s) is/are are incorrect ?
- (A) The order is 1
- (B) The order is 0.5
- (C) For  $[A_0] = 1\text{M}$ ,  $t_{1/2} = 100\sqrt{10}$  sec
- (D) For  $[A_0] = 1\text{M}$ ,  $t_{1/2} = 100$  sec
19.  $\text{PCl}_5$  dissociations into  $\text{PCl}_3$  and  $\text{Cl}_2$  thus  $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ .
- If the total pressure of the system in equilibrium is P at a density  $\rho$  and temperature T, M is the relative molar mass of  $\text{PCl}_5$  and if the vapour density of the gas mixture at equilibrium has the value of 62 when the temperature is  $230^\circ\text{C}$ , then the correct statements are :
- (A)  $\alpha = \frac{PM}{PRT} - 1$
- (B)  $\alpha = \frac{PRT}{PM} - 1$
- (C)  $\frac{P}{\rho} = 0.3327 \text{ atm gm/L}$
- (D)  $\frac{P}{\rho} = 2.327 \text{ atm gm/L}$
20. Which of the following is/are lyophobic colloids ?
- (A) Gold sol
- (B)  $\text{As}_2\text{S}_3$  sol
- (C) Starch sol
- (D)  $\text{Fe}(\text{OH})_3$  sol
21. Which of the following is/are not true for lyophobic colloid ?
- (A) These are prepared by special indirect methods
- (B) The particles must travel towards the anode or cathode under the influence of an electric field
- (C) These are called an intrinsic colloid
- (D) Small quantity of electrolyte is sufficient to cause precipitation of these

(Numerical Value type) :

22. 11 moles  $\text{N}_2$  and 12 moles of  $\text{H}_2$  mixture reacted in 20 litre vessel at 800 K. After equilibrium was established, 6 mole of  $\text{H}_2$  was present. 3.58 ltr of liquid water is injected in equilibrium mixture and resultant gaseous mixture suddenly cooled to 300K. If the value of final pressure is x atm. Find the value of 2x. Neglect vapour pressure of liquid solution. Assume
- (i) all  $\text{NH}_3$  dissolved in water,

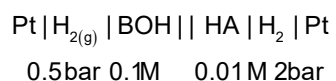
(ii) no change in volume of liquid and

(iii) no reaction of  $N_2$  and  $H_2$  at 300 K.



23. For the just coagulation of 250 ml of  $Fe(OH)_3$  sol, 2 ml of 1 M  $Na_2SO_4$  electrolyte is required. What is the coagulating value of  $Na_2SO_4$  electrolyte?

24. Calculate EMF (in V) of the following cell



Given  $2.303 \frac{RT}{F} = 0.06$

$\log 2 = 0.30$

$K_a (HA) = 10^{-6}$

$K_b (BOH) = 10^{-5}$

## Answer key

|     |       |     |       |     |       |     |       |     |      |     |      |
|-----|-------|-----|-------|-----|-------|-----|-------|-----|------|-----|------|
| 1.  | (D)   | 2.  | (C)   | 3.  | (A)   | 4.  | (B)   | 5.  | (A)  | 6.  | (D)  |
| 7.  | (D)   | 8.  | (C)   | 9.  | (B)   | 10. | (C)   | 11. | (D)  | 12. | (C)  |
| 13. | (ABD) | 14. | (ABD) | 15. | (AD)  | 16. | (BCD) | 17. | (AC) | 18. | (AD) |
| 19. | (AC)  | 20. | (ABD) | 21. | (ABD) | 22. | (45)  | 23. | (8)  | 24. | 0.40 |